Code No: 133AJ **R16**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech II Year I Semester Examinations, September/October - 2023
DIGITAL LOGIC DESIGN
(Common to CSE, IT)

Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

PART – A

(25 Marks)

1.a) Convert (10110)2 to Gray code and (110101)G to binary number. [2]
b) What is Floating point number representation and give examples? [3]
c) What are universal gates and why they are called as universal gates? [2]
d) Draw a four variable Karnaugh map and List its simplification methods. [3]
e) What is a multiplexer? What is the function of a multiplexer's select input? [2]
f) Can more than one decoder output be activated at one time? Justify your answer? [3]
g) What is a flip-flop? Write the logic diagram of J-K flip-flop List all the pins. [2]
h) Write the difference between synchronous and asynchronous sequential circuits. [3]
i) What are shift micro operations and what are the different types. [2]
j) Explain about sequential programmable logic devices. [3]

PART – B

(50 Marks)

2.a) Convert the following i) $(53.625)_{10}$ to (? $)_2$ ii) $(3FD)_{16}$ to (? $)_2$ iii) $(A69.8)_{16}$ to (? $)_{10}$
b) Perform the decimal subtraction in 8-4-2-1 BCD using 9's complement
i) Subtract 79 from 26 ii) Subtract 748 from 983. [6+4]

OR

3.a) Explain in detail about Weighted, non weighted, Error minimizing and, error detecting and error correcting codes with examples.
b) Check whether the received code 10101100 is correctly received or not if even parity is used. [6+4]

4.a) Convert the given Boolean function into standard sum of Minterms form. F = x'y+y'z+xz.
b) Simplify the following Boolean function using K-Map method in (i) SOP form (ii) POS form $F(A, B, C, D)= \Sigma(0, 6, 8, 13, 14); \Sigma d (2, 4,10)$ [4+6]

OR

5.a) Reduce using mapping the following expression and implement the real minimal expression in Universal logic. F= $\sum$m (0, 2, 4, 6, 7, 8, 10, 12, 13, 15)
b) Reduce the following expression using k-map f = π M (1, 4, 5, 11, 12, 14) .d (6, 7, 15).[5+5]

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6.a) Explain the full adder and Full Subtractor with the help of a neat logic diagram and truth table.

b) Design a 4:16 decoder with basic gates with control inputs and enable inputs. [5+5]

OR

7.a) Draw and explain the 8×1 MUX. and 3/8 decoder.

b) Explain with neat sketches the working of 4 bit Priority encoder and a 4 bit magnitude comparator. [5+5]

8.a) What is Race condition and explain about the operation of clocked RS flip flop and D flip-flop.

b) Design a 4 bit shift left register and shift right using D flip-flops flip flops. [5+5]

OR

9.a) Write the differences between synchronous and Asynchronous Counters.

b) Explain the operation of the 4-bit asynchronous counter with a J-K flip-flops and draw its state diagram. [4+6]

10.a) Explain the logic implementation of a 32 × 4 bit ROM using decoder of a suitable size.

b) What do you mean by register transfer? Explain in detail. Also discuss three state bus buffer. [5+5]

OR

11.a) With a neat diagram explain in detail about two dimensional memory decoding scheme.

b) A combinational circuit is defined by the functions:

F1= m (3, 5, 7) F2 = m(4,5,7)

Implement the circuit with a PLA having 3 inputs, 3 product terms and two outputs. [4+6]

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